

1. Let $H_n : \frac{x^2}{1+n} - \frac{y^2}{3+n} = 1$, $n \in \mathbb{N}$. Let k be the smallest even value of n such that the eccentricity of H_k is a rational number. If l is the length of the latus rectum of H_k , then $21l$ is equal to
[2023 (11 Apr Shift 1)]
2. Let the tangent to the parabola $y^2 = 12x$ at the point $(3, \alpha)$ be perpendicular to the line $2x + 2y = 3$. Then the square of distance of the point $(6, -4)$ from the normal to the hyperbola $\alpha^2 x^2 - 9y^2 = 9\alpha^2$ at its point $(\alpha - 1, \alpha + 2)$ is equal to
[2023 (11 Apr Shift 2)]
3. Let m_1 and m_2 be the slopes of the tangents drawn from the point $P(4, 1)$ to the hyperbola $H : \frac{y^2}{25} - \frac{x^2}{16} = 1$. If Q is the point from which the tangents drawn to H have slopes $|m_1|$ and $|m_2|$ and they make positive intercepts α and β on the x -axis, then $\frac{(PQ)^2}{\alpha\beta}$ is equal to
[2023 (13 Apr Shift 1)]
4. The foci of a hyperbola are $(\pm 2, 0)$ and its eccentricity is $\frac{3}{2}$. A tangent, perpendicular to the line $2x + 3y = 6$, is drawn at a point in the first quadrant on the hyperbola. If the intercepts made by the tangent on the x - and y -axes are a and b respectively, then $|6a| + |5b|$ is equal to
[2023 (13 Apr Shift 2)]

ANSWER KEYS

1. (306)

2. (116)

3. (8)

4. (12)

1. (306)

Given,

Equation of hyperbola,

$$H_n : \frac{x^2}{1+n} - \frac{y^2}{3+n} = 1, n \in \mathbb{N}$$

Now using the eccentricity formula we get,

$$e^2 = 1 + \frac{3+n}{1+n}$$

$$\Rightarrow e^2 = \frac{2n+4}{n+1} = \frac{2(n+2)}{n+1}$$

Now check when $(n+1) = 9, 25, 49, \dots$ is perfect square and then at the same time $2(n+2)$ should also be perfect square,

So checking for $n = 8 \rightarrow e^2 = \frac{20}{9}$

$$n = 24 \rightarrow e^2 = \frac{52}{25}$$

$$n = 48 \rightarrow e^2 = \frac{100}{49} \Rightarrow e = \frac{10}{7}$$

Hence, for $n = 48$ both $n+1$ & $2(n+2)$ are perfect square,

Now we know that,

$$\text{Length of latusrectum is given by } l = \frac{2b^2}{a} = \frac{2(n+3)}{\sqrt{n+1}} = \frac{2 \times 51}{7}$$

$$\text{Hence, } 21l = \frac{42 \times 51}{7} = 306$$

2. (116)

Given,

The tangent to the parabola $y^2 = 12x$ at the point $(3, \alpha)$ be perpendicular to the line $2x + 2y = 3$

So, Slope of tangent $= 1 = \frac{6}{\alpha} \Rightarrow \alpha = 6$

So, equation of hyperbola will be,

$$36x^2 - 9y^2 = 324$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{36} = 1$$

Now equation of tangent at $(5, 8)$ will be,

$$\frac{5x}{9} - \frac{8y}{36} = 1$$

$$\Rightarrow 5x - 2y = 9$$

So, slope of normal $= \frac{-2}{5}$

Now finding, equation of normal $y - 8 = \frac{-2}{5}(x - 5)$

$$\Rightarrow 5y - 40 = -2x + 10$$

$$\Rightarrow 5y + 2x = 50$$

Now finding, distance from $(6, -4)$, we get,

$$= \left| \frac{12 - 20 - 50}{\sqrt{29}} \right| = \frac{58}{\sqrt{29}}$$

$$= 2\sqrt{29}$$

$$\text{Hence, } (2\sqrt{29})^2 = 116$$

3. (8)

Given that $H : \frac{y^2}{25} - \frac{x^2}{16} = 1$

Equation of tangent to the hyperbola is: $y = mx + \sqrt{25 - 16m^2}$

Passes through $(4, 1)$

$$\Rightarrow y = mx + \sqrt{25 - 16m^2}$$

$$\Rightarrow 1 = 4m + \sqrt{25 - 16m^2}$$

$$\Rightarrow 4m^2 - m - 3$$

$$\Rightarrow m_1 = 1, m_2 = -\frac{3}{4}$$

$$\Rightarrow |m_1| = 1, |m_2| = \frac{3}{4}$$

Equation of tangents with slopes 1 & $\frac{3}{4}$

$$\Rightarrow y = x - 3 \dots (i)$$

$$\Rightarrow y = \frac{3}{4}x - 4 \dots (ii)$$

$$(i) \& (ii) \Rightarrow Q \equiv (-4, -7)$$

$$P \equiv (4, 1)$$

$$\Rightarrow (PQ)^2 = 8^2 + 8^2 = 128$$

On converting (i) and (ii) in intercept form we get,

$$(i) \Rightarrow \alpha = 3, (ii) \Rightarrow \beta = \frac{16}{3}$$

$$\Rightarrow \frac{(PQ)^2}{\alpha\beta} = \frac{128}{3 \times \frac{16}{3}} = 8$$

Hence this is the required option.

4. (12)

The given equation of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, ae = 2, e = \frac{3}{2}$

$$\Rightarrow a = \frac{4}{3}$$

$$\Rightarrow b^2 = a^2 e^2 - a^2 = 4 - \frac{16}{9} = \frac{20}{9}$$

The tangent is perpendicular to $2x + 3y = 6$.

$$\text{Slope of tangent is } m = \frac{-1}{-\frac{2}{3}} = \frac{3}{2}$$

$$\text{Equation of tangent, } m = \frac{3}{2}$$

$$\text{Equation of tangent is } y = mx - \sqrt{a^2 m^2 - b^2}$$

$$\Rightarrow y = \frac{3}{2}x - \sqrt{\frac{16}{9} \times \frac{9}{4} - \frac{20}{9}}$$

($\because$ Tangent is in 1st quadrant $\Rightarrow C < 0$)

$$\Rightarrow y = \frac{3}{2}x - \frac{4}{3}$$

Converting the equation into intercept form:

$$\Rightarrow \frac{x}{(\frac{8}{9})} + \frac{y}{(-\frac{4}{3})} = 1$$

$$\Rightarrow |6a| + |5b| = \left| \frac{6 \times 8}{9} \right| + \left| \frac{5 \times (-4)}{3} \right| = 12$$

Hence this is the required option.